



HK IMO Training 2021-2022
Problem Set 7
For 13 November 2021



Problem 25. Let $n \geq 2$, and let $a_1, a_2, \dots, a_n$ be real numbers such that $a_j > 1$ for $j = 1, 2, \dots, n$, and $|a_j - a_{j+1}| < 1$ for $j = 1, 2, \dots, n-1$. Prove that

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} < 2n - 1.$$

Problem 26. Let p be an odd prime. How many tuples $(x_1, x_2, \dots, x_p)$ are there with $x_j \in \{0, 1, \dots, p-1\}$ for each j , such that $p \mid x_1x_2 + x_2x_3 + x_3x_4 + \dots + x_{p-1}x_p + x_px_1$ and $p \nmid x_1 + x_2 + \dots + x_p$?

Problem 27. There are $n \geq 2$ segments inside a unit circle with centre O such that the total length of the segments is $2\sqrt{n}$. Prove that there exists a circle with centre O that meets at least two of the segments (possibly meeting at the endpoint).

Problem 28. Let $S = \{1, 2, \dots, n\}$, and let $\mathcal{F}$ be a family of subsets of S . It is known that for any $A, B \in \mathcal{F}$, we have $A \cup B \in \mathcal{F}$. Also, for any $A \in \mathcal{F}$, we have $3 \nmid |A|$. Prove that there exist $x, y \in S$ such that every A in $\mathcal{F}$ contains at least one of x and y .